

June 6, 2025

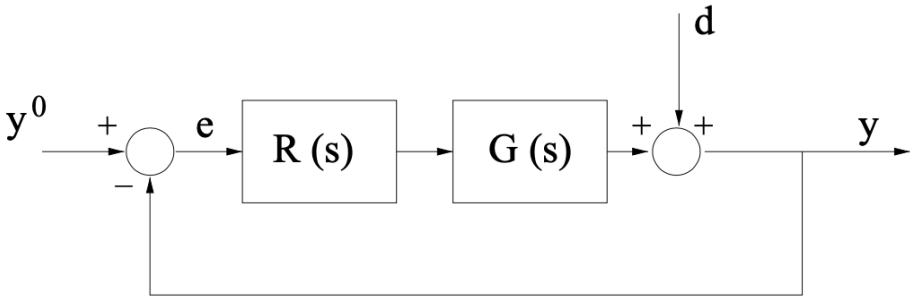
Design of a Proper Controller for a Plant with a Parameter Affected by Interval Uncertainty

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Description of the Feedback Control System

Consider the feedback control system described by the block-scheme shown in the following figure



where $G(s)$ is given by

$$G(s) = 10 \frac{(s + 1)(s + 200)}{(s^2 + 20Bs + 100)(s + 250)}, \quad B \in [0.005, 1.0]$$

Note: This means that the system has a transfer function with two complex conjugate poles, the natural pulsation of which is known exactly, but the exact damping $\bar{B}$ is not known. All that is known of the effective B damping (which is unknown) is that it takes on a value contained in the assigned range $[0.005, 1.0]$.

Question

Q1: Design a **proper controller** $C(s)$ such that all following requirements are simultaneously met:

- Zero steady-state error in unit step response (closed-loop) $e(\infty) = 0$, against disturbances given by

$$\begin{cases} y^o(t) &= 1(t) \\ d(t) &= A \cdot 1(t), A \in \mathbb{R} \end{cases} \implies e_\infty = 0$$

where $1(t)$ denotes the unit step function;

- phase margin greater than or equal to 30° : $\varphi_m \geq 30^\circ$;
- settling time [at 1% as usual] in the (closed-loop) unit step output less than or equal to 30 seconds: $t_{s\,1\%} \leq 30 \text{ sec.}$

Hint

How to handle the unknown damping factor B ? A good practice is to identify the **worst-case scenario** and design the controller $C(s)$ according to that scenario.

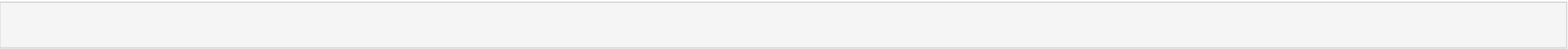
Solution

Initialisation code (clearing the MATLAB workspace, adding the path of the BodeDiagram subfolder to the MATLAB search-path)

```
clear variables

% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

% let's define the transfer function builder element
s=tf('s');
```



Preliminary Analysis

$$G(s) = 10 \frac{(s + 1)(s + 200)}{(s^2 + 20Bs + 100)(s + 250)}, \quad B \in [0.005, 1.0]$$

According to (Q1.1), the regulator must have a pole at $s = 0$: $C_1(s) = \frac{\mu_C}{s}$, $\mu_C > 0$

```
dampMIN = 0.005; dampMAX = 1.0;
dampR = 0.5;

% static regulator, according to (Q1,1)
muC = 1;
C1s = muC/s;

% what is the worst-case scenario?
Gs1 = 10*(s+1)*(s+200)/(s^2+20*dampMIN*s+100)/(s+250);
Gs2 = 10*(s+1)*(s+200)/(s^2+20*dampMAX*s+100)/(s+250);
Gs3 = 10*(s+1)*(s+200)/(s^2+20*dampR*s+100)/(s+250);

Ls1 = C1s*Gs1;
Ls2 = C1s*Gs2;
Ls3 = C1s*Gs3;
```

Let's analyse the (Q1.3) requirement:

$$t_{s1\%} \leq 10 \iff \begin{cases} \varphi_m > 75^\circ \implies t_{s1\%} = \frac{4.6}{\omega_c} \leq 30 \implies \omega_c \geq \frac{46}{300} \\ \varphi_m < 75^\circ \implies t_{s1\%} = \frac{4.6}{\xi \omega_n} \approx \frac{4.6}{\frac{\varphi_m}{100} \omega_c} \leq 30 \implies \varphi_m \cdot \omega_c \geq \frac{46}{3} \end{cases} \quad (*)$$

Consider the first scenario ($\varphi_m > 75^\circ$):

```
% the Bode diagrams

Bcolors = [0 0.4470 0.7410; ...
           0.9290 0.6940 0.1250; ...
           0.4940 0.1840 0.5560; ...
           0.4660 0.6740 0.1880; ...
           0.3010 0.7450 0.9330; ...
           0.8500 0.3250 0.0980; ...
           0.6350 0.0780 0.1840]; % some different colors

hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
```

```

drawBodediagrams(Ls2, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +75; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                        % to guarantee the feasibility of the design

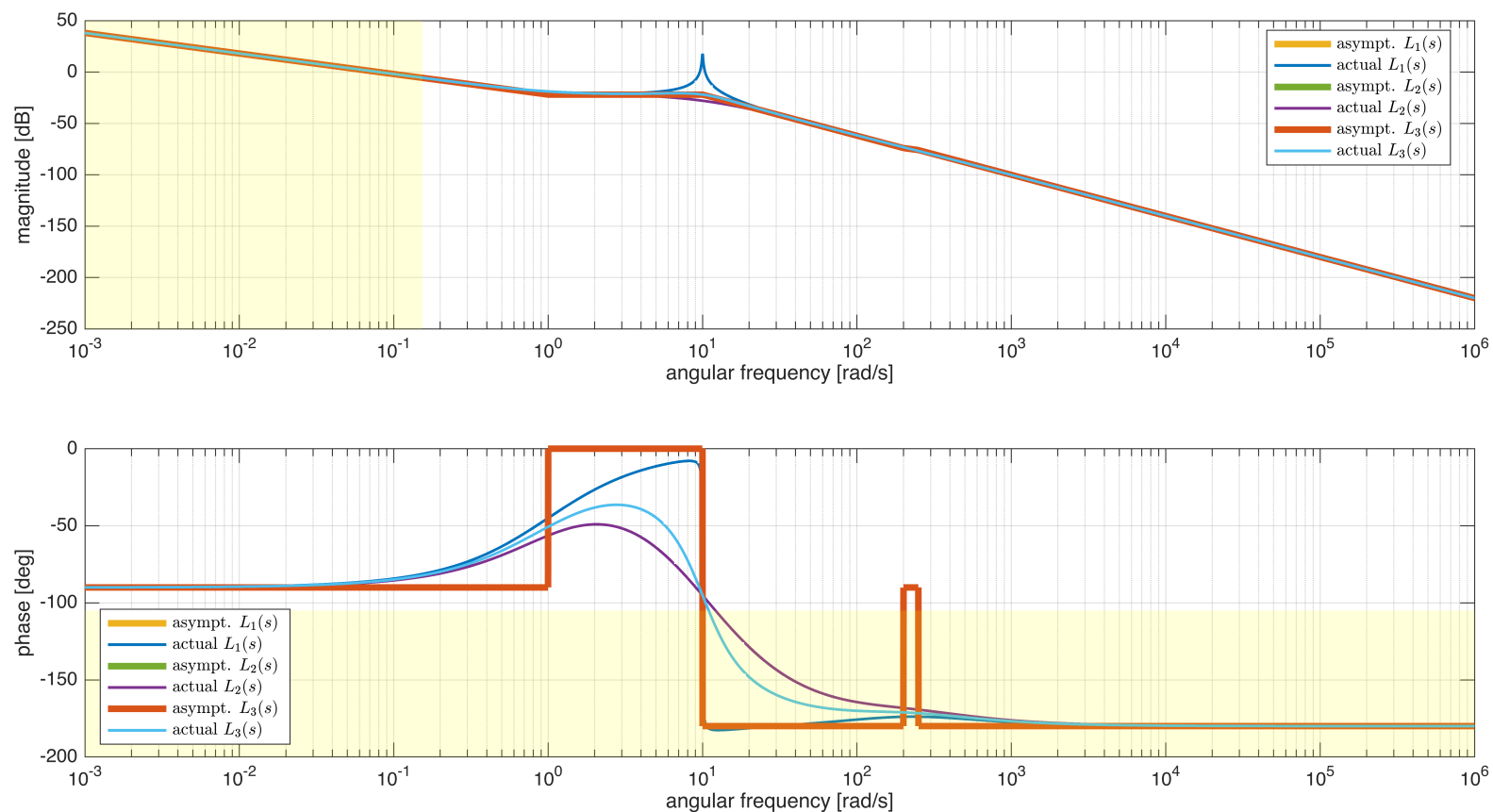
YLIMS = ylim(ha2);
OMLIMS = xlim(ha2);
% the (Q1.2) constraint
minOmega1 = OMLIMS(1);
maxOmega1 = OMLIMS(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 , YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1 2 3 4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor','none'); % the pale yellow region is the forbidden one

omC1 = 46/300; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);
OMLIMS = xlim(ha1);
minOmega1 = OMLIMS(1);
maxOmega1 = OMLIMS(2);
Vert2 = [minOmega1 , YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1 2 3 4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor','none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','southwest')

```



The design problem seems feasible, with the simple controller $C_1(s) = \frac{\mu_C}{s}$ if we plan to obtain the crossover angular frequency in the interval $\omega_c \in [0.2, 2.0]$ rad/sec.

The worst-case scenario corresponds to the highest value for the damping coefficient B , i.e., $B = 1.0$.

With the aim to ease the design, let's insert a term with a zero in the controller, with the zero at $z = -10$

```

C2s = (1+s/10);
Cs = C1s*C2s;

Ls1NEW = Cs*Gs1;
Ls2NEW = Cs*Gs2;
Ls3NEW = Cs*Gs3;

hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1NEW, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2NEW, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3NEW, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +75; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                        % to guarantee the feasibility of the design

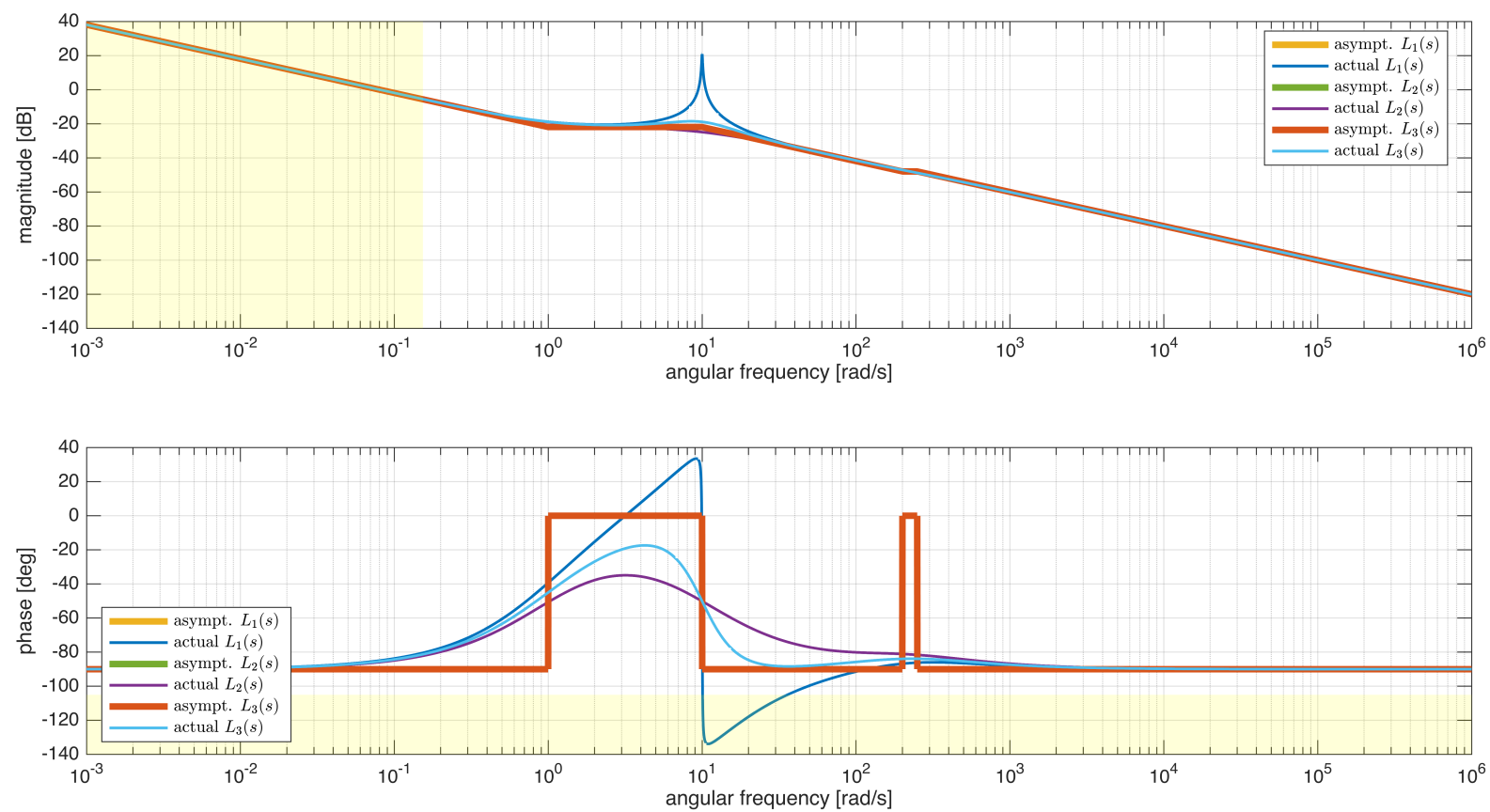
YLIMS = ylim(ha2);
OMLIMS = xlim(ha2);
% the (Q1.2) constraint
minOmega1 = OMLIMS(1);
maxOmega1 = OMLIMS(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 , YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1 2 3 4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor', 'none'); % the pale yellow region is the forbidden one

omC1 = 46/300; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);
OMLIMS = xlim(ha1);
minOmega1 = OMLIMS(1);
maxOmega1 = OMLIMS(2);
Vert2 = [minOmega1 , YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1 2 3 4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor', 'none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
      'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
      'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
      'Interpreter', 'latex', 'Location', 'best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
      'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
      'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
      'Interpreter', 'latex', 'Location', 'southwest')

```



The design problem seems feasible, with the controller $C(s) = C_1(s) \cdot C_2(s)$ if we plan to obtain the crossover angular frequency in the interval $\omega_c \in [0.2, 9.0]$ rad/sec or $\omega_c > 200$ rad/sec.

The worst-case scenario corresponds to the highest value for the damping coefficient B , i.e., $B = 1.0$.

An even better design scenario corresponds to the dynamic controller $C_2(s) = \frac{\left(1 + \frac{s}{10}\right)^2}{\left(1 + \frac{s}{200}\right)}$

$$C(s) = \mu_C \frac{\left(1 + \frac{s}{10}\right)^2}{s \left(1 + \frac{s}{200}\right)} \Rightarrow L(s) = 20 \mu_C \frac{(s+1)(s+10)^2}{s(s^2 + 20Bs + 100)(s+250)}, \quad B \in [0.005, 1.0]$$

```

C2s = (1+s/10)^2/(1+s/200);
Cs = minreal(C1s*C2s);

Ls1NEW = Cs*Gs1;
Ls2NEW = Cs*Gs2;
Ls3NEW = Cs*Gs3;

hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1NEW, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2NEW, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3NEW, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +75; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                        % to guarantee the feasibility of the design

ylim(ha2, [-180, 180]);
YLIMS = ylim(ha2);
OMLIMS = xlim(ha2);
% the (Q1.2) constraint
minOmega1 = OMLIMS(1);
maxOmega1 = OMLIMS(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 , YLIMS(1); minOmega1 ,PhLIM; ...
        maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1 2 3 4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor','none'); % the pale yellow region is the forbidden one

omC1 = 46/300; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);

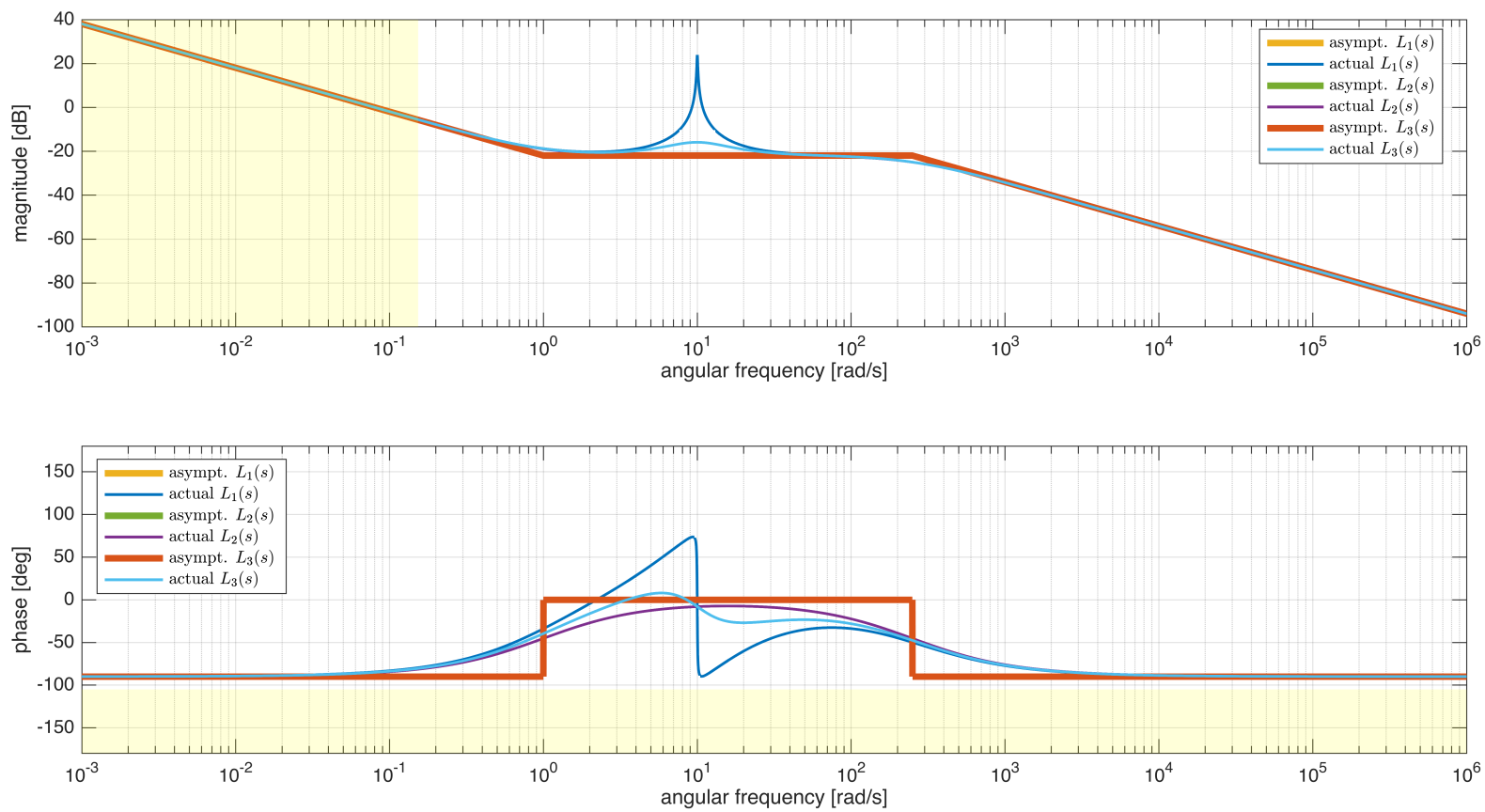
```

```

OMLIMs = xlim(ha1);
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
Vert2 = [minOmega1 , YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1 2 3 4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
      'FaceColor','yellow','FaceAlpha',.15, ...
      'EdgeColor','none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','northwest')

```



In this scenario, we can guarantee the phase margin $\bar{\varphi}_m = +90^\circ$ when $\omega_c \geq 0.2$ rad/sec.

What about the second scenario in Eq. (*)? Consider the minimum allowable value for the phase margin:

$$\varphi_m = 30^\circ \implies \omega_c \geq \frac{46}{90} \approx 0.512 \text{ rad/sec}$$

```

hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +30; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                        % to guarantee the feasibility of the design

YLIMS = ylim(ha2);
OMLIMs = xlim(ha2);
% the (Q1.2) constraint
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 , YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1 2 3 4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...

```

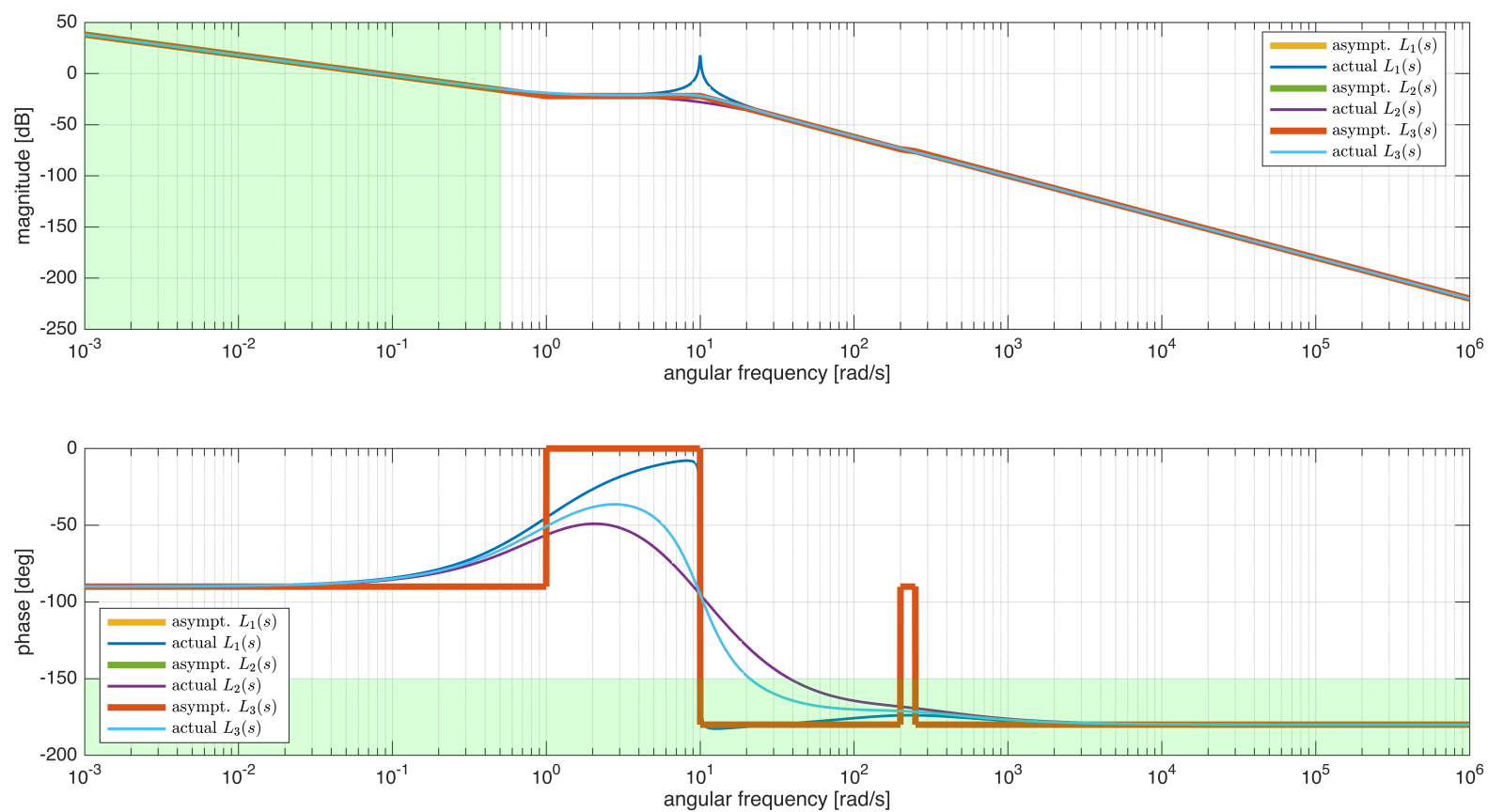
```

'FaceColor','green','FaceAlpha',.15, ...
'EdgeColor','none'); % the pale yellow region is the forbidden one

omC1 = 0.512; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);
OMLIMs = xlim(ha1);
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
Vert2 = [minOmega1 , YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1 2 3 4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
      'FaceColor','green','FaceAlpha',.15, ...
      'EdgeColor','none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
        'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
        'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
        'Interpreter','latex','Location','southwest')

```



Also this time, the design problem is feasible, with the simple controller $C_1(s) = \frac{\mu_C}{s}$ if we plan to obtain the crossover angular frequency in the interval $\omega_c \leq 5.0$ rad/sec.

The worst-case scenario corresponds to the highest value for the damping coefficient B , i.e., $B = 1.0$.

Tuning the Controller

According to the previously performed analysis, the controller

$$C(s) = \mu_C \frac{\left(1 + \frac{s}{10}\right)^2}{s \left(1 + \frac{s}{200}\right)}$$

allows the fulfilment of all the requirements, with the phase margin $\bar{\varphi}_m = +90^\circ$ when $\omega_c \geq 0.2$ rad/sec.

Let's tune the gain constant μ_C with the aim to obtain $\bar{\omega} = 10.0$ rad/sec as crossover frequency in the worst-case scenario:

```
barOMc1 = 1j*10;
```

```
mag_L2s_bar0Mc1 = abs(freqresp(Ls2NEW, bar0Mc1));
muC_VAL = 1/mag_L2s_bar0Mc1
```

```
muC_VAL =
12.4479
```

Verify the performance of the controller $C_1(s)$ in every scenario

```
Cs = muC_VAL*Cs;

Ls1FINAL = Cs*Gs1;
Ls2FINAL = Cs*Gs2;
Ls3FINAL = Cs*Gs3;

[gm, pm, o_gm, o_pm] = margin(Ls1FINAL)
```

```
gm =
Inf
pm =
147.3499
o_gm =
NaN
o_pm =
69.5526
```

```
[gm2, pm2, o_gm2, o_pm2] = margin(Ls2FINAL)
```

```
gm2 =
Inf
pm2 =
171.9989
o_gm2 =
NaN
o_pm2 =
10.0002
```

```
[gm3, pm3, o_gm3, o_pm3] = margin(Ls3FINAL)
```

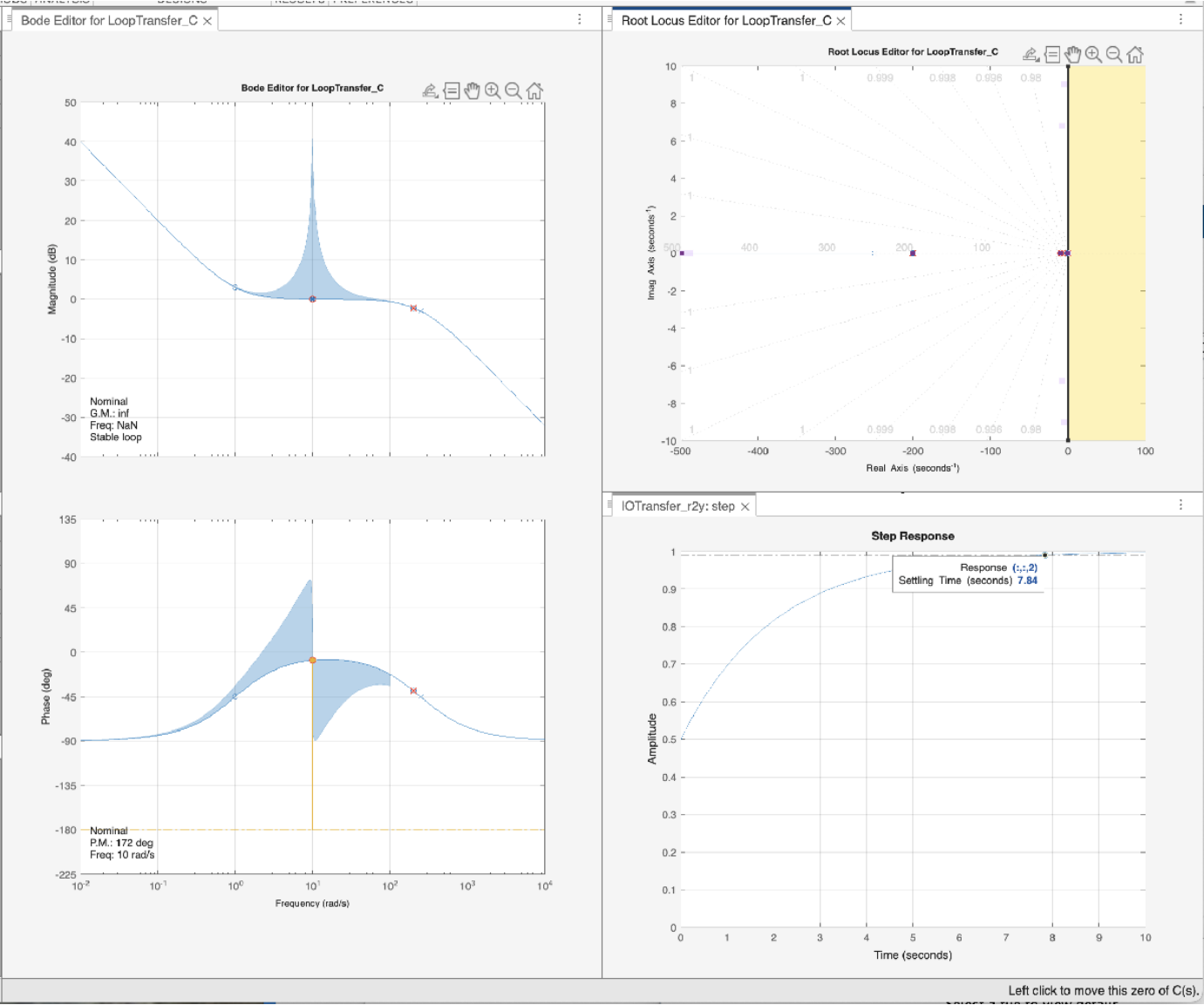
```
gm3 =
Inf
pm3 =
156.0696
o_gm3 =
NaN
o_pm3 =
64.1841
```

In conclusion, the controller $C(s)$ allows to fulfil every requirement.

Fine Tuning the Controller using the controlSystemDesigner App

```
% let's create a multi-model structure
Gs = stack(1, Gs1, Gs2, Gs3);

controlSystemDesigner(Gs, Cs)
```



The corresponding controlSystemDesigner session file is `multimodel_solution_2025_06_06_FA_gr_C_ControlSystemDesignerSession.mat`